

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027**

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	2026-27 (ODD)	Course Name	Lab: Hybrid Computing Systems
Semester	7	Course Code	23IOT1704
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	07
Course Outcome	1. Understand Cloud Computing architecture, service models, deployment models, and virtualization. 2. Apply cloud platforms, storage services, and containerization for application deployment. 3. Implement Edge and Fog Computing solutions for IoT applications. 4. Analyze and develop integrated Cloud-Fog-Edge computing applications based on performance evaluation.
Aim	Data Processing at the Edge using MQTT and Node-RED.
Theory (100 words)	MQTT is a lightweight messaging protocol designed for communication between devices with limited processing power and network resources. It is widely used in IoT applications because it requires low bandwidth and supports reliable message exchange. MQTT communication is based on a publish-subscribe mechanism, where a device publishes messages to a specific topic and other devices subscribe to that topic to receive the messages. An MQTT Broker acts as an intermediate server that receives messages from publishers and distributes them to the appropriate subscribers. This allows IoT devices to communicate without directly connecting to each other. Node-RED is a visual, flow-based development tool that allows users to create applications by connecting different functional nodes. It can receive MQTT messages, process sensor information, apply conditions or calculations, and display the resulting data. In an edge computing environment, Node-RED can operate on a local computer or edge device. Sensor information received through MQTT can

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	therefore be processed locally instead of continuously sending raw data to a remote cloud server. For example, Node-RED can check whether a temperature value exceeds a specified limit and generate an alert immediately. The basic communication flow is: IoT Device → MQTT Publisher → MQTT Broker → Node-RED → Data Processing → Output Node-RED can perform various operations such as data filtering, value conversion, threshold checking, data formatting, and visualization. Combining MQTT with Node-RED provides an efficient approach for developing IoT applications that require local processing, low communication delay, reduced bandwidth consumption, and quick responses.
Procedure and Execution (100 Words)	1. Install Node.js on the Windows computer. 2. Open the Node.js Command Prompt and install Node-RED using: npm install -g --unsafe-perm node-red 3. Launch Node-RED by executing: node-red 4. Open a web browser and access the Node-RED editor. 5. Insert an Inject node to simulate IoT sensor data. 6. Add a Function node to perform local processing on the received sensor value. 7. Place a Debug node to display the processed output. 8. Connect the nodes in the following sequence: Inject → Function → Debug 9. Configure the Inject node with a sample temperature value, such as 30°C. 10. Program the Function node with a condition to determine whether the temperature is Normal or High. 11. Select Deploy to activate the configured Node-RED flow. 12. Press the Inject button and check the processed result in the Debug panel. 13. Modify the temperature value and perform the test again with different readings.

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14. Confirm that the sensor information is processed locally by Node-RED at the edge before any cloud-based processing.

Stepwise Screenshots with steps:

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sheik>node --version
v22.19.0

C:\Users\sheik>npm --version
11.12.1

C:\Users\sheik>npm install -g --unsafe-perm node-red
npm warn "node-red" is being parsed as a normal command line argument.
npm warn Unknown cli config "--unsafe-perm". This will stop working in the next major version of npm.

added 13 packages, removed 12 packages, and changed 264 packages in 22s

75 packages are looking for funding
  run 'npm fund' for details
npm notice
npm notice New minor version of npm available! 11.12.1 -> 11.19.1
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.19.1
npm notice To update run: npm install -g npm@11.19.1
npm notice

C:\Users\sheik>node-red --version
Node-RED v5.0.7
Node.js v22.19.0
Windows_NT 10.0.26200 x64 LE

C:\Users\sheik>

C:\Users\sheik>node-red
19 Sep 12:03:09 - [info]
Welcome to Node-RED
=====

19 Sep 12:03:09 - [info] Node-RED version: v5.0.7
19 Sep 12:03:09 - [info] Node.js version: v22.19.0
19 Sep 12:03:09 - [info] Windows_NT 10.0.26200 x64 LE
19 Sep 12:03:10 - [info] Loading palette nodes
19 Sep 12:03:11 - [info] Settings file : C:\Users\sheik\.node-red\settings.js
19 Sep 12:03:11 - [info] Context store : 'default' [module=memory]
19 Sep 12:03:11 - [info] User directory : \Users\sheik\.node-red
19 Sep 12:03:11 - [warn] Projects disabled : editorTheme.projects.enabled=false
19 Sep 12:03:11 - [info] Flows file : \Users\sheik\.node-red\flows.json
19 Sep 12:03:11 - [info] Server now running at http://127.0.0.1:1880/
19 Sep 12:03:11 - [warn]

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Your flow credentials file is encrypted using a system-generated key.

If the system-generated key is lost for any reason, your credentials
file will not be recoverable, you will have to delete it and re-enter
your credentials.

You should set your own key using the 'credentialSecret' option in
your settings file. Node-RED will then re-encrypt your credentials
file using your chosen key the next time you deploy a change.
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```



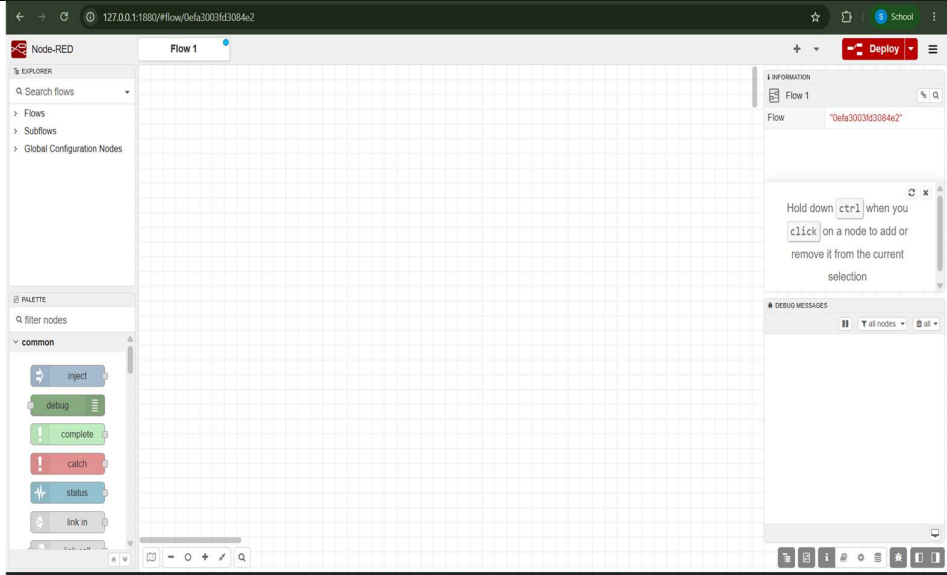
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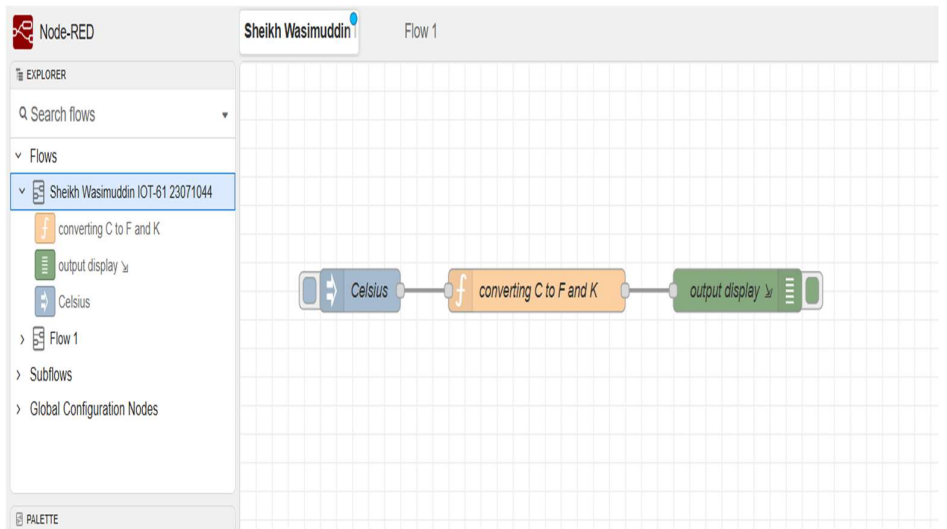
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The top screenshot shows the Node-RED web interface. The main workspace is empty. The left sidebar contains an 'EXPLORER' panel with a search bar and a list of flows and subflows. Below it is a 'PALETTE' panel with a search bar and a list of common nodes including inject, debug, complete, catch, status, and link in. The right sidebar contains an 'INFORMATION' panel showing details about the current flow and a 'DEBUG MESSAGES' panel.



The bottom screenshot shows the Node-RED web interface with a flow named 'Sheikh Wasimuddin' selected. The flow consists of three nodes connected in a sequence: a 'Celsius' node, a 'converting C to F and K' node, and an 'output display' node. The left sidebar shows the 'EXPLORER' panel with the selected flow highlighted. The right sidebar shows the 'INFORMATION' panel and the 'DEBUG MESSAGES' panel.



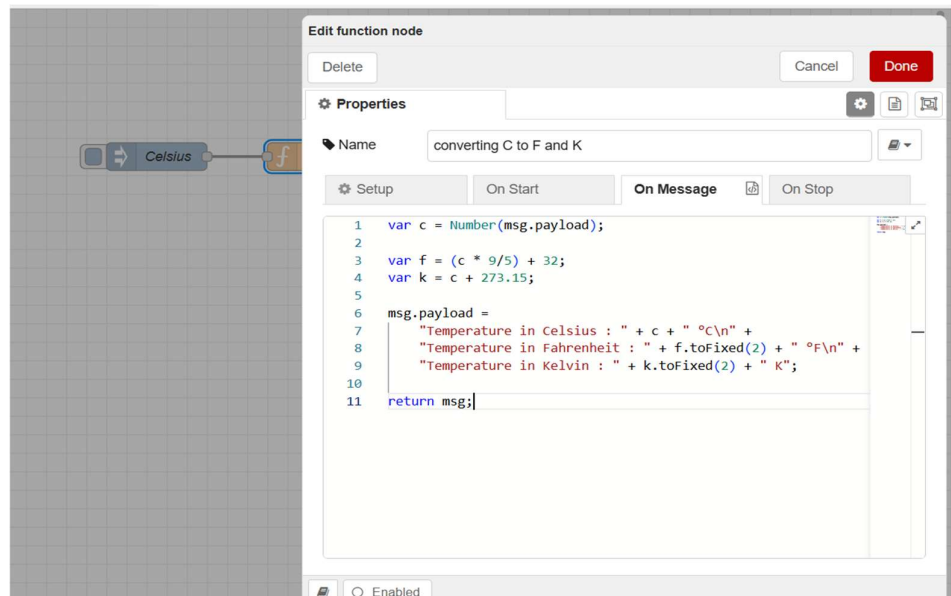
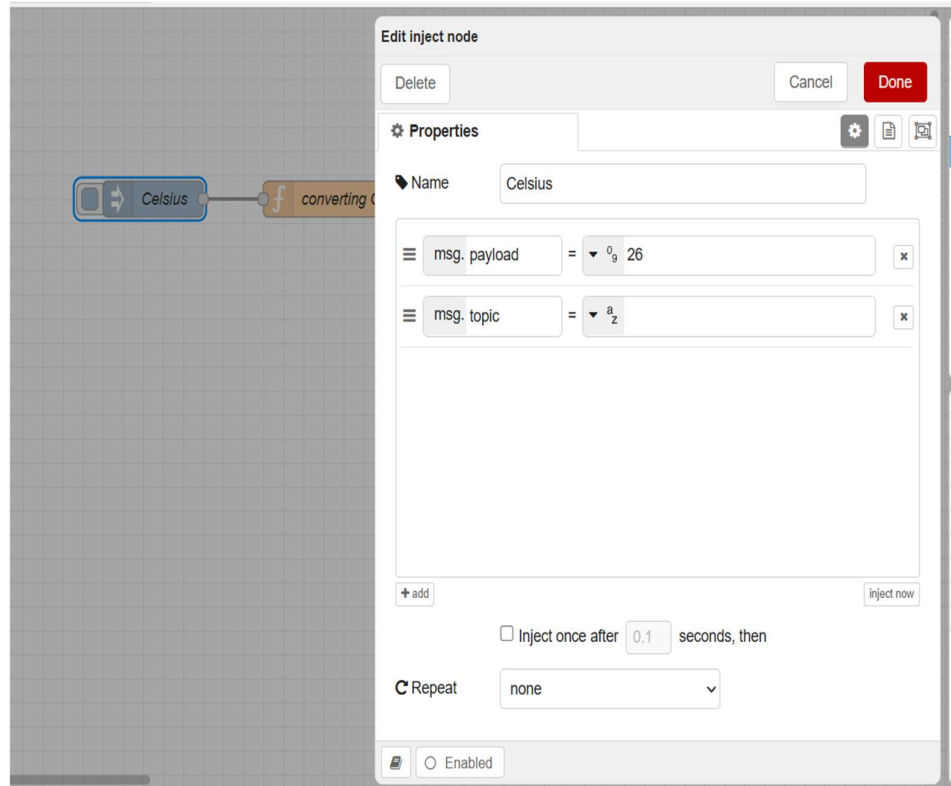
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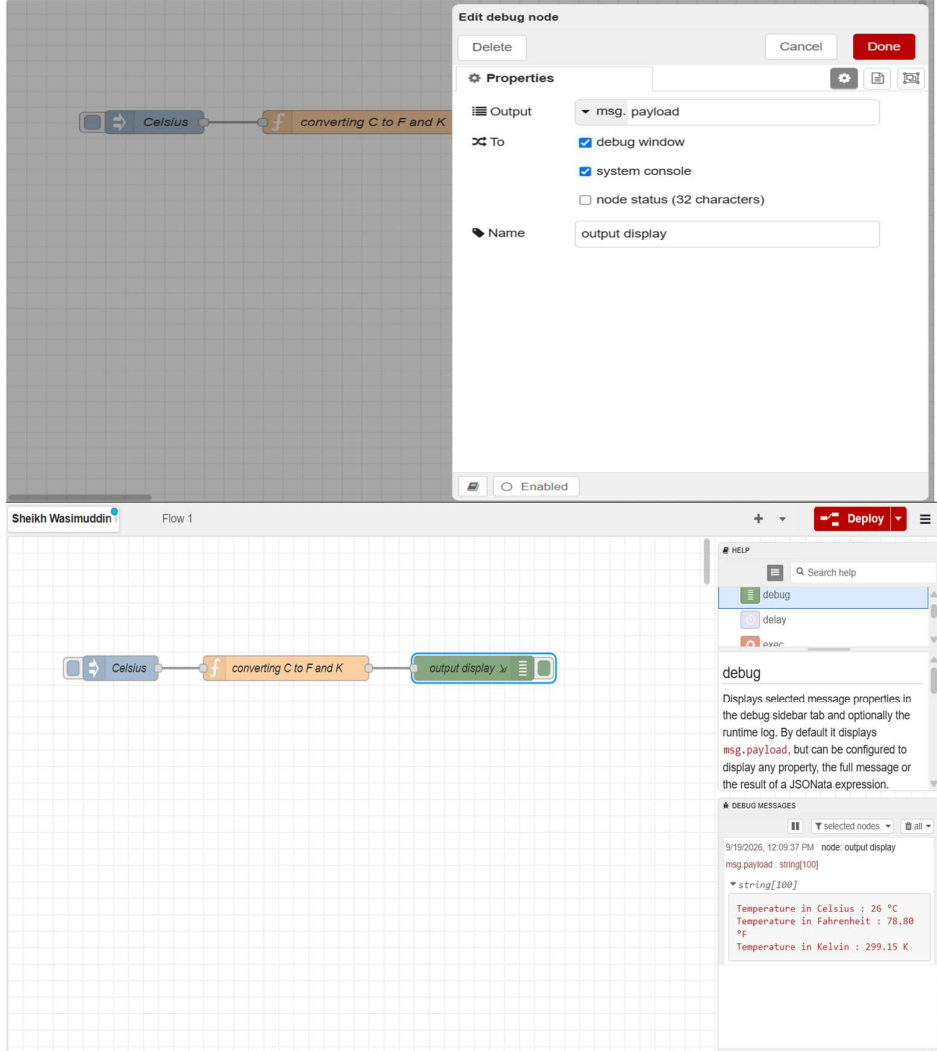
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Conclusion	The Node-RED flow was successfully created and executed for local IoT data processing. A Celsius temperature value was generated using the Inject node and processed by the Function node to calculate its Fahrenheit and Kelvin equivalents. The converted values were successfully displayed through the Debug node. Thus, the practical demonstrated how Node-RED can perform real-time data processing at the edge using a simple flow-based approach.
Date	25/09/2026